SRI: N01LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 1$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{1} F_{s,d} + O(\mathrm{SR}^2)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{1} F_{t,d} + O(\mathrm{SR}^2)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

2 Consistency relations

2.1 $n = 0$

For $p = 1$, the absolute cutoff is $W = 2$:

$$\alpha_0^t = \sum_{d=1}^{2} R_{0,d} + O(\mathrm{SR}^3).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

2.2 $n = 1$

For $p = 1$, the absolute cutoff is $W = 3$:

$$\alpha_1^t = \sum_{d=2}^{3} R_{1,d} + O(\mathrm{SR}^4).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

2.3 $n = 2$

For $p = 1$, the absolute cutoff is $W = 4$:

$$\alpha_2^t = \sum_{d=3}^{4} R_{2,d} + O(\mathrm{SR}^5).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = &-0.00390625r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{2,4} = &\, 0.0013427734375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.125\left(\alpha_0^s\right)^2 r^2 \\ &+ 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s\alpha_1^s r \end{aligned}$$

2.4 $n = 3$

For $p = 1$, the absolute cutoff is $W = 5$:

$$\alpha_3^t = \sum_{d=4}^{5} R_{3,d} + O(\mathrm{SR}^6).$$

Absolute order 4.

$$\begin{aligned} R_{3,4} = &\, 0.00146484375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.0234375\alpha_0^s r^3 \\ &+ 0.109375\left(\alpha_0^s\right)^2 r^2 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s\alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{3,5} = &-0.000762939453125r^5 + 0.125\alpha_3^s r \\ &+ 0.0419921875\alpha_1^s r^3 - 0.125\left(\alpha_0^s\right)^4 r \\ &- 0.375\left(\alpha_1^s\right)^2 r - 0.078125\alpha_2^s r^2 \\ &- 0.28125\left(\alpha_0^s\right)^3 r^2 - 0.01708984375\alpha_0^s r^4 \\ &- 0.1201171875\left(\alpha_0^s\right)^2 r^3 + 0.75\left(\alpha_0^s\right)^2\alpha_1^s r \\ &+ 0.4375\alpha_0^s\alpha_1^s r^2 - 0.5\alpha_0^s\alpha_2^s r \end{aligned}$$